

PROJECT FLOW – VIRO-AI (FULL USER-CENTRIC FRONTEND FLOW)

1. HERO PAGE

User lands on hero/marketing page with main CTA “Get Started” and “Explore Demo”. Shows 3D protein-antidote teaser image. Nav: Features, Pricing, Docs, Login.

2. SIGNUP / LOGIN

Signup: Name, Email, Password, Org Name. Login: email/password or SSO. On success → fetch profile with GET /auth/me and redirect to Explore.

3. EXPLORE PAGE

Shows all projects: cards with title, created_at, status. Filters for My Projects, Demo, Status filters. Right side: “Create Project” button and “Quick Analysis” card with pretrained datasets. Clicking Create Project opens full creation page.

4. CREATE PROJECT PAGE

Sections:

- Project Meta: Title, Description, Tags, Project Type.
- Required Files: PDB (structure), CSV (assay data), FASTA (sequence). Each validated. Upload via presigned URL.
- Options: Processing profile, sensitivity, timezone.
- QUICK ANALYSIS card at bottom: same UI; lists pretrained datasets. Selecting dataset allows instant analysis without uploads.

Button: Create Project → POST /projects. Redirect to results page instantly.

5. PROJECT CREATION BEHAVIOR

Backend returns project_id + status=processing + estimated_wait ~10s. Redirect to Results page. Results may load instantly if cached/light; else show skeleton viewer + spinner for ≤10 seconds.

6. RESULTS PAGE (CORE)

Route: /projects/{id}/results

Layout:

- Left: toolbar: Back, Share, Download, Mark Completed/Failed, Re-run.
- Center: 3D viewer (GLB) with rotate/zoom, overlays.
- Overview Card: status, KPIs (binding affinity, efficacy, confidence, mutation count, drug candidate count, modification count).
- Tabs on Right: Overview (default), Mutations, Drug Candidates, Modifications.
- Activity log at bottom/right.

6.1 INSTANT/DELAY BEHAVIOR

If processing done: instantly show full overview + 3D viewer.

If not: show skeleton & toast "Analysis starting — results will appear within ~10s". Poll `/projects/{id}/status` every 2s.

7. DETAILED PANELS

7.1 MUTATIONS PANEL (9 fields)

Each record includes:

1. Residue ID (chain+number)
2. Original AA
3. Mutated AA
4. Position
5. Occurrence frequency
6. Predicted impact score
7. Structural context
8. Sequence conservation
9. Notes/suggestions

7.2 DRUG CANDIDATES PANEL (11 fields)

1. Candidate ID/Name
2. Molecular formula/SMILES
3. Binding affinity
4. Predicted efficacy score
5. Confidence
6. Off-targets
7. ADMET summary
8. Synthesis feasibility
9. Binding site list
10. Supporting evidence
11. Recommendation/next step

7.3 MODIFICATIONS PANEL (11 fields)

1. Modification ID
2. Type
3. Target residues
4. Proposed change

5. Effect on binding
6. Effect on stability
7. Immunogenicity
8. Ease of implementation
9. Suggested protocol
10. Dependencies
11. Confidence/provenance

8. HISTORY PAGE

List all projects with filters. Show status (processing/completed/failed). Status is user-controlled. Scientists update status manually depending on lab outcomes.

9. STATUS LOGIC

Mark Completed or Mark Failed with reason. PATCH /projects/{id} updates status and logs in audit trail.

10. DASHBOARD / USER STATS

KPIs:

- Total projects
- Completed rate
- Avg processing time
- Most used pretrained datasets
- Recent activity

Clickable to filter projects.

11. SETTINGS

User-level:

- Notifications
- Default processing profile
- Timezone
- API keys

Org-level:

- Retention policy
- Upload limits
- Role/access control

12. PROFILE PAGE

Top right avatar. Shows user info, orgs, role, editable fields.

13. QUICK ANALYSIS BEHAVIOR

Clicking a pretrained dataset:

- Prefills project creation
- Runs analysis instantly ($\leq 5s$)
- Saved as project with `quick_analysis=true`
- Same results page workflow

14. API SUMMARY

Key endpoints used by frontend:

- POST `/auth/login`
- GET `/auth/me`
- GET `/projects`
- POST `/projects/presign`
- POST `/projects`
- GET `/projects/{id}/status`
- GET `/projects/{id}/results`
- PATCH `/projects/{id}`
- GET `/datasets/pretrained`

15. RESULTS PAYLOAD SHAPE (SUMMARY)

Contains: overview, viewer_asset (GLB), mutations[], drug_candidates[], modifications[], activity log.

16. UX RULES

- Results appear instantly OR within 10 seconds max.
- Panels load lazily.
- Clicking 3D residue highlights mutation.
- Drug candidate click shows 3D pose overlay.
- Modifications have “Add to experiment plan”.

17. ERROR STATES

- Invalid files → inline error.
- Processing fail → retry or contact support.

- Timeout >10s → detailed progress loader.

18. ACCEPTANCE CRITERIA

- Create Project redirects within 2s.
- Results appear instantly or within 10s.
- All panels show required fields.
- Quick Analysis works with pretrained datasets.
- Status changes persist.
- History updates instantly.
- Viewer loads GLB and interactions work.

17. DEVELOPER HANDOFF

Deliver mockups, full API schemas, sample PDB/CSV/FASTA, pretrained dataset list, E2E test flows.